Math 54 — Week 5

Solutions

P1.

Idea. Swap to get a leading 1, use replacements only, count swaps.

1. $R_1 \leftrightarrow R_2$ (one swap): rows $(1,3,0,2)$, $(2,4,-2,6)$, $(-1,-1,4,1)$, $(3,8,1,5)$.
2. $R_2 - 2R_1$, $R_3 + R_1$, $R_4 - 3R_1$: $(0,-2,-2,2)$, $(0,2,4,3)$, $(0,-1,1,-1)$.
3. $R_3 + R_2$: $(0,0,2,5)$. $R_4 - \frac{1}{2}R_2$: $(0,0,2,-2)$.
4. $R_4 - R_3$: $(0,0,0,-7)$. Diagonal: $1,-2,2,-7$, product 28.
5. One swap: $\det = -28$.

P2. $\det(AB) = 3(-2) = -6$. $\det(A^{-1}) = \frac{1}{3}$. $\det(A^T) = 3$. $\det(2A) = 2^3 \cdot 3 = 24$.

$$\det(B^T A^{-1} B) = (-2) \cdot \tfrac{1}{3} \cdot (-2) = \tfrac{4}{3}.$$

P3.

Idea. Invertible exactly when $\det \neq 0$.

Expand along row 1: $1(15-1) - 2(6-0) + h(2-0) = 14 - 12 + 2h = 2(h+1)$. Invertible exactly when $h \neq -1$.

P4. $\det A = 5 - 6 = -1$.

$$\det A_1(\mathbf{b}) = \det \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix} = -7, \qquad \det A_2(\mathbf{b}) = \det \begin{bmatrix} 5 & 1 \\ 3 & 4 \end{bmatrix} = 17.$$

So $x_1 = 7$, $x_2 = -17$. Check: $35 - 34 = 1$, $21 - 17 = 4$.

P5.

Idea. Move one vertex to the origin; the two adjacent sides are the columns.

Subtract $(-1,0)$: the vertices become $(0,0)$, $(1,5)$, $(2,-4)$, $(3,1)$, and $(1,5) + (2,-4) = (3,1)$, so the sides are $(1,5)$ and $(2,-4)$. The area is

$$\left| \det \begin{bmatrix} 1 & 2 \\ 5 & -4 \end{bmatrix} \right| = |-14| = 14.$$

The transformation multiplies areas by $|\det \left[\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right]| = 5$, so the image has area $5 \cdot 14 = 70$.

P6.

1. (a) Yes: it is $\mathrm{Nul} \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}$.
2. (b) No: $(1,0)$ and $(0,-1)$ are in it, but their sum $(1,-1)$ has $ab = -1 < 0$.
3. (c) No: the zero polynomial has value 0 at 0, not 1.
4. (d) Yes: it is $\mathrm{Span}\{(1,1,0),(-2,0,3)\}$.
5. (e) Yes: $0^T = 0$; $(A+B)^T = A^T + B^T = A + B$; $(cA)^T = cA^T = cA$.

P7.

$$\left[\begin{array}{ccc|c} 1 & -3 & -4 & 3 \\ -4 & 6 & -2 & 3 \\ -3 & 7 & 6 & -4 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & 5 & -9/2 \\ 0 & 1 & 3 & -5/2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Consistent, so $\mathbf{b} \in \operatorname{Col} A$. The coefficient part gives $x_1 = -5x_3$, $x_2 = -3x_3$: $\operatorname{Nul} A = \operatorname{Span}\{(-5, -3, 1)\}$.

P8.

$$A\mathbf{u} = (6 - 8 + 2 + 0,\ -6 + 10 - 7 + 0,\ 9 - 14 + 8 + 0) = (0, -3, 3) \neq \mathbf{0}:$$

$\mathbf{u} \notin \operatorname{Nul} A$. Row reducing $[\,A \ \ \mathbf{v}\,]$ gives pivots in columns 1, 2, 4 only (not in the last column), so it is consistent: $\mathbf{v} \in \operatorname{Col} A$. $\mathbf{u}$ has 4 entries and $\operatorname{Col} A \subseteq \mathbb{R}^3$, so $\mathbf{u}$ cannot be in $\operatorname{Col} A$.